



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SELVA VINAYAGA FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

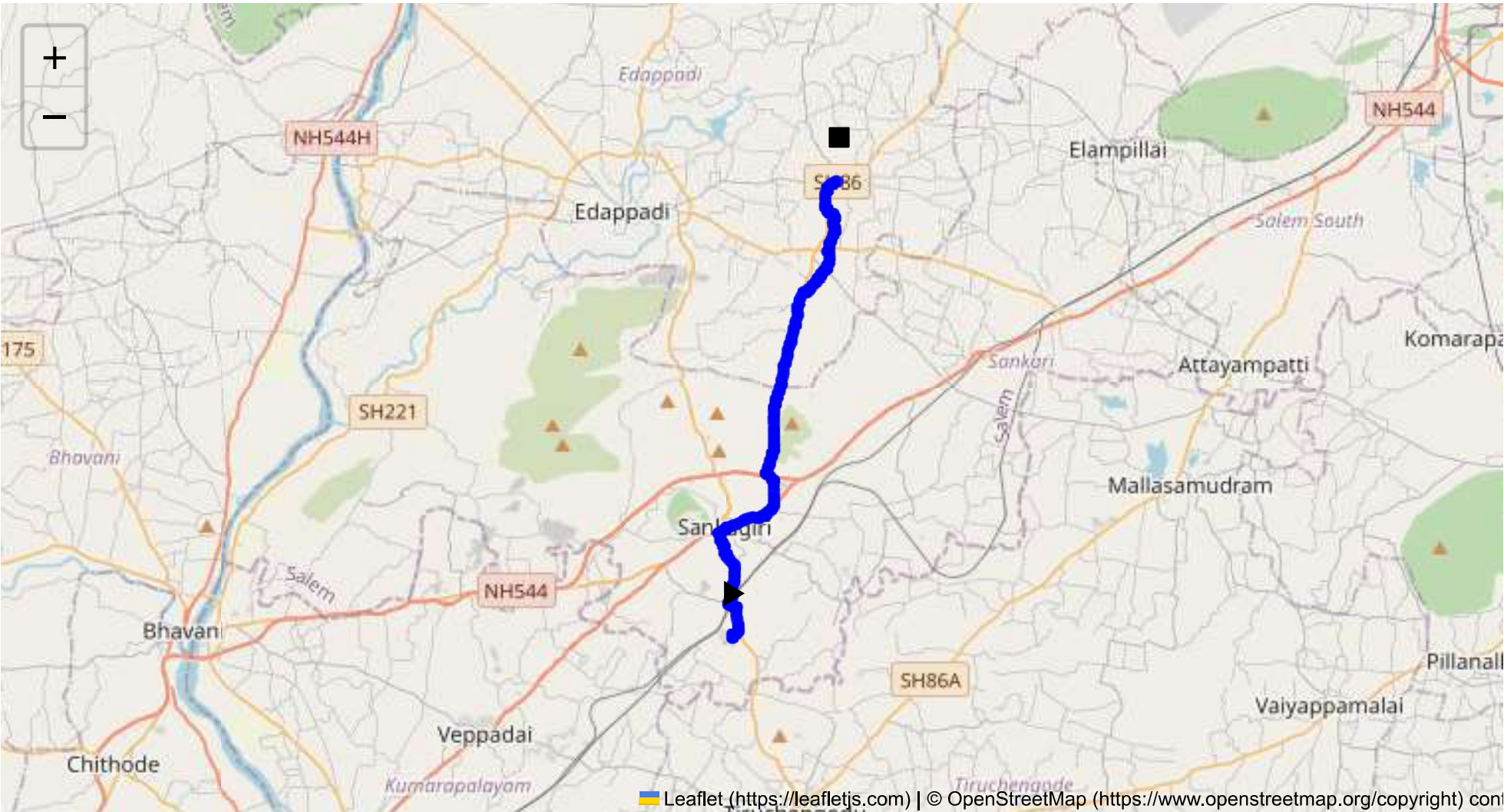
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 20.37 km
Estimated Duration: 0.6 hours
Adjusted Duration (Heavy Vehicle): 0.7 hours
Start: (11.4381, 77.8734)
End: (11.593695, 77.909306)



Welcome to the Journey Risk Management Study

Route Overview

The route from CVQF+23W in Sangagiri to HWV5+7PQ in Moolapathai, Tamil Nadu, covers approximately 20.37 kilometers and typically takes about 34 minutes for heavy vehicles transporting hazardous materials. It likely involves regional roads with a mix of village paths and state highways.

1. Overview of the Route Map

The route primarily navigates through local roads that connect villages and might merge with state highways. This type of route usually involves navigating through rural areas, potentially passing through small communities or towns.

2. Weather Conditions and Hazards

Tamil Nadu experiences a tropical climate. The key weather-related hazards may include:

- **Monsoon Season:** Heavy rainfall from June to September, which can cause flooding and slippery roads.
- **Summer Heat:** High temperatures from March to May, which can affect vehicle performance and driver fatigue.
- Periodic cyclones might also affect the region, leading to road closures and severe disruptions.

3. Traffic Patterns

Traffic is generally lighter in rural areas, but periodic congestion might occur:

- **Peak Hours:** Typically morning (8-10 AM) and evening (5-7 PM) are the busier times, as local residents commute for work and school.
- **Congestion Areas:** Intersection points with state highways or around small town centers may experience delays.

4. Road Quality and Infrastructure

- **Rural Roads:** These might be narrow with limited infrastructure support, posing challenges for heavy vehicles.
- **State Highways:** Expected to be in better condition but can still have rough patches and limited shoulders.
- **Potholes and Unpaved Sections:** Particularly after monsoon rains, these can be frequent.

5. Alternative Routes for Emergencies

- In case of roadblocks or severe congestion, diverting to parallel state highways, or using bypass roads through larger towns where wider roads and better facilities exist, could be considered.

6. Local Regulations for Hazardous Material Transport

- Indian regulations require proper documentation and labeling for hazardous material transport.
- Movement of hazardous materials may be restricted during peak traffic hours or night time in populated areas to reduce risks.

7. Historical Incidents

- While specific incident reports are not available, rural roads in India sometimes witness accidents involving heavy vehicles due to overloading or speeding.
- Awareness of local accident hotspots as identified by regional traffic departments would be beneficial.

8. Environmental Considerations

- **Sensitive Areas:** Forest patches or areas close to water bodies are present, warranting careful driving to prevent ecological disturbances.
- Awareness of local wildlife crossings is also necessary.

9. Communication Coverage

- Major towns and sections of highways generally have good mobile coverage.
- Rural areas and stretches between towns might experience dead zones with unreliable network service.

10. Emergency Response Times

- **Urban Areas:** Faster responses, typically within 20-30 minutes, due to closer proximity to emergency services.
- **Rural Areas:** May take significantly longer, up to 60 minutes or more, due to distance from towns and medical facilities.

11. Overall Summary of Risk Assessment

While generally navigable, this route presents moderate risks due to weather impacts, road conditions, and potentially erratic traffic patterns. Heavy vehicles carrying hazardous materials need to exercise caution particularly during adverse weather conditions and in congested areas. Regular updates on road conditions and weather forecasts will be crucial in mitigating risks. Backup plans for emergencies and clear communication with local authorities for fire, medical, and police assistance are essential for a safe journey.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43811, 77.87348	15 KM/Hr
1	Turn	Medium	11.43956, 77.87340	30 KM/Hr
2	Turn	Medium	11.43971, 77.87348	30 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
4	Turn	High	11.45007, 77.87201	15 KM/Hr
5	Turn	High	11.47407, 77.86839	15 KM/Hr
6	Turn	Medium	11.48363, 77.88780	30 KM/Hr
7	Turn	Medium	11.48446, 77.88787	30 KM/Hr
8	Turn	Medium	11.49451, 77.88489	30 KM/Hr
9	Turn	High	11.55614, 77.89764	15 KM/Hr
10	Turn	High	11.58317, 77.90647	15 KM/Hr
11	Turn	Medium	11.59061, 77.90583	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
1	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
2	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
3	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
4	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
6	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
8	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
9	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
5	school	Goverment Boys Higher Secondary School	11.477659, 77.8659632	30 km/h	Medium
7	marketplace	Sunday Market	11.4796502, 77.8708458	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45007, 77.87201



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.47407, 77.86839



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.48363, 77.88780



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.48446, 77.88787



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49451, 77.88489



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.55614, 77.89764



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.58317, 77.90647



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59061, 77.90583

